# Document Control

- System Name: Application

# Executive Summary

- This BRD details the ingestion, cleansing, enrichment, join, SCD Type 2 history, aggregation, and pipeline implementation of customer and order data using Delta Live Tables.
- The system "Application" transforms CSV business data into historical and aggregated Delta tables, ensuring traceability and data quality from source to customer analytics.

# Business Objectives

- Load customer and order source CSVs into enterprise-grade Delta tables.
- Cleanse and deduplicate data for reliability.
- Enrich order data with computed TotalAmount.
- Create a historical join (ordersummary) with SCD Type 2 handling.
- Aggregate daily customer spend.
- Implement a repeatable, automated data pipeline using Delta Live Tables.

# In-Scope

- Source CSV ingestion into Delta tables.
- Creation and management of customer, order, ordersummary, customeraggregatespend tables.
- Data cleansing (remove Null literals and duplicates).
- Computed column (TotalAmount).
- Join and SCD Type 2 history mechanisms.
- Aggregation for spend analytics.
- Implementation in Delta Live Tables.

# Out-of-Scope

- Integration of non-specified data sources or additional fields.
- Security, access controls, SLAs, monitoring, and alerting.
- Performance tuning or optimizations.
- Scheduling, orchestration, or DevOps outside Delta Live Tables.
- User interfaces, reports, or analytics layers.

# Stakeholders

- Information not present in requirements input.

# Business Requirements

### BR-APP-001 — Source Data Ingestion

- Description: Ingest customer and order CSV data into Delta tables from specified filesystem paths.
- Business Value: Enables structured data foundation for all downstream processing.
- Priority: High

### BR-APP-002 — Data Cleansing

- Description: Remove "Null"/Null literals and duplicate records from both customer and order tables.
- Business Value: Prevents data quality issues and improves accuracy.
- Priority: High

### BR-APP-003 — Order Enrichment

- Description: Compute TotalAmount for each order as PricePerUnit × Qty.
- Business Value: Supports downstream analytics and reporting.
- Priority: High

### BR-APP-004 — Historical Join with SCD Type 2

- Description: Join customer and order on CustId to create ordersummary with full history tracking (SCD Type 2).
- Business Value: Maintains historical accuracy and supports compliance reporting.
- Priority: High

### BR-APP-005 — Aggregated Analytics

- Description: Aggregate ordersummary by Name and Date into customeraggregatespend with sum(TotalAmount).
- Business Value: Delivers ready-to-use business insights for spend analysis.
- Priority: High

### BR-APP-006 — Table and Schema Management

- Description: Ensure all Delta tables (customer, order, ordersummary, customeraggregatespend) are created and maintained as specified.
- Business Value: Guarantees consistent schema and lineage for audit and traceability.
- Priority: Medium

### BR-APP-007 — Delta Live Tables Implementation

- Description: Execute the end-to-end business pipeline with Delta Live Tables, as directed.
- Business Value: Drives automation, scalability, and operational reliability.
- Priority: High

# Assumptions & Constraints

- No business logic or attributes beyond the explicit requirements shall be added.
- Implementation is strictly governed by Delta Live Tables.
- StartDate, EndDate, and Active fields for SCD Type 2 are referenced but definitions are absent.

# Success Metrics

- 100% of valid CSV records successfully ingested and cleansed.
- Computed TotalAmount available for all order records.
- All customer changes tracked with SCD Type 2 accuracy.
- Aggregated customer spend tables are accurate and replayable.
- End-to-end pipeline reproducible via Delta Live Tables without manual intervention.

# Risks & Mitigations

- Incomplete SCD Type 2 field specifications may delay implementation — Mitigate by clarifying with stakeholders.
- Data quality issues beyond literal "Null" and duplicates unaddressed — Add further validation if future requirements permit.
- Error handling, monitoring, scheduling, and security are not covered — Future extensions to scope may address these.

# Approval

- Approval information not present in requirements input.

# Appendix

Appendix A: Source Definitions

| Source Name | Type | Location | Format | Notes |
|---|---|---|---|---|
| Customer source | CSV | /Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/customerdata | CSV | Loaded into Delta table customer |
| Order source | CSV | /Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/orderdata | CSV | Loaded into Delta table order |

Appendix B: Target Definitions

| | Layer | Table | Columns | Notes |
|---|---|---|---|---|
| | Bronze | customer | CustId, Name, EmailId, Region | Ingested |
| | Bronze | order | OrderId, ItemName, PricePerUnit, Qty, Date, CustId | Ingested |
| | Silver | ordersummary | CustId, Name, EmailId, Region, OrderId, ItemName, PricePerUnit, Qty, Date | Joined ar |
| atespend | Gold | customeraggregatespend | Name, TotalAmount, Date | Aggregat |

Appendix C: Transformations

| Step | Transformation | Input | Output | Logic |
|---|---|---|---|---|
| 1 | Load CSV | customerdata/orderdata | customer/order | Ingest raw data |
| 2 | Remove nulls/duplicates | customer/order | customer/order | Data cleansing |
| 3 | Compute TotalAmount | order | order | TotalAmount = PricePerUnit × Qty |
| 4 | Join with SCD2 | customer + order | ordersummary | SCD Type 2 join on CustId |
| 5 | Aggregate spend | ordersummary | customeraggregatespend | sum(TotalAmount) by Name, Date |

Appendix D: Business Rules

| Rule ID | Area | Description | Source |
|---|---|---|---|
| BR-001 | Computation | TotalAmount = PricePerUnit × Qty | Requirements |
| BR-002 | Cleansing | Remove "Null"/Null and duplicates | Requirements |
| BR-003 | Join | customer join order on CustId | Requirements |
| BR-004 | SCD Type 2 | Maintain SCD Type 2 record history | Requirements |
| BR-005 | Aggregation | Aggregate sum(TotalAmount) per Name/Date | Requirements |

Appendix E: Process Flow

| Step | Description | Dependency |
|---|---|---|
| 1 | Ingest source CSV to Delta | None |
| 2 | Cleanse data | Step 1 |
| 3 | Compute TotalAmount on orders | Step 2 |
| 4 | Join and build ordersummary with SCD Type 2 | Step 3 |
| 5 | Aggregate spend into customeraggregatespend | Step 4 |

Appendix F: Architecture Diagram (Graphviz)

- The following presents the process flow and lineage of data objects described above as a data engineering pipeline.

```
digraph D {

subgraph cluster_ingest {

label = "Ingestion";

"Customer Source CSV" -> "Customer Delta Table";

"Order Source CSV" -> "Order Delta Table";

}
```

```
subgraph cluster_cleansing {

label = "Cleansing";

"Customer Delta Table" -> "Customer Cleansed";

"Order Delta Table" -> "Order Cleansed";

}

subgraph cluster_enrichment {

label = "Enrichment";

"Order Cleansed" -> "Order with TotalAmount";

}

subgraph cluster_joining {

label = "Historical Join";

"Customer Cleansed" -> "Ordersummary SCD Type 2";

"Order with TotalAmount" -> "Ordersummary SCD Type 2";

}

subgraph cluster_aggregation {

label = "Aggregation";

"Ordersummary SCD Type 2" -> "Customer Aggregate Spend";

}

}
```